

LOUIS BERTHIER

PhD Candidate in Applied Mathematics and Machine Learning.

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EDUCATION

École Polytechnique | PhD in Applied Mathematics and Machine Learning 2024 - Present
Paris, France

- Industrial Thesis (CIFRE) in collaboration with Michelin, supervised by [Pr. Eric Moulines](#) & [Dr. Ahmed Shokry](#).
- Developing adaptive and conformal soft sensors for real-time quality prediction in multistage industrial processes.

Imperial College London | MSc Advanced Computing in Machine Learning 2022 - 2023
London, United Kingdom

- Diploma with merit (2:1), and thesis with distinction (1st).
- Research internship at the AIRL, under the supervision of [Pr. Antoine Cully](#).
- Relevant courses: Machine & Deep Learning, Reinforcement Learning, Computer Vision, Natural Language Processing, Mathematics for Machine Learning, Computational Finance, Robot Learning.

IMT Mines Alès | Diplôme d'Ingénieur - MSc Artificial Intelligence & Data Science 2020 - 2023
Alès, France

- First class honours with jury's congratulations
- Ranked 2nd in CS & AI department, top 5% of the class
- Research internship at the CNRS, under the supervision of [Dr. Ludovic Gardy](#), & [Pr. Christophe Hurter](#).
- Relevant courses: Advanced Probabilities & Statistics, Operational Research, Algorithms & Complexity, Machine & Deep Learning, Software Engineering

WORK EXPERIENCE

Augmodo | Computer Vision Scientist Aug 2025 - Present
Seattle, Washington, United States

Improving open world computer vision pipelines for real-world deployments in retail using uncertainty quantification. In collaboration with [Sacha Hu](#), [Ashley Napier](#), & [Shubham Wagh](#).

- Building uncertainty-aware open-world classifiers for product identification, QC automation, and empty-shelf area detection (RGB+depth) using test-time augmentation to quantify prediction confidence, selectively accepting only high-confidence outputs to reduce manual review workload at 100k+ images/day.
- Fine-tuning object detectors (YOLO and the transformer-based RF-DETR) for empty-shelf detection, leveraging hard-negative examples to push detection performance while preserving other shelf-object classes, with training and fine-tuning on cloud GPUs.
- Designing automated relabeling pipelines to update legacy bounding-box annotations at scale, adapting large existing datasets to newer model architectures without manual re-annotation.

Michelin | ML Research Engineer (PhD Contract) Mar 2024 - Present
Clermont-Ferrand, France

Developing conformal and adaptive soft sensors for real-time quality prediction in industrial batch processes. Supervised by [Guillaume Ramelet](#) & [Dr. Maxime Moreaud](#).

- Created and open-sourced [TorchSOM](#) (Apache 2.0, 90% test coverage), the reference PyTorch library for Self-Organizing Maps: 74–98% faster training than MiniSom on CPU and the lowest topographic error in every benchmark (34–81% below MiniSom); it backs the similarity-based retrieval across the soft-sensing and conformal-prediction pipelines.
- Designed and benchmarked five adaptive soft-sensing strategies on 35,125 production batches; an online SHAP-based feature selection cut 167 process variables to about 18 (about 90%), and the best hybrid reduced prediction error by about 59% short-term and over 90% under year-long drift.
- Developed *Mask-Valid Conformal Prediction*: binary monitoring masks that interpolate between marginal and anytime coverage with restart-validity, for sequential batch monitoring without past test labels.
- Developed *Self-Organized Conformal Prediction*: unsupervised group discovery with a Self-Organizing Map and local calibration, closing regional coverage gaps with near cell-conditional guarantees and no model retraining.

London, United Kingdom

Model-based uncertainty quantification for quality-diversity optimization in robotic environments.

Supervised by [Pr. Antoine Cully](#).

- Enhanced MAP-Elites sample efficiency and noise robustness by integrating probabilistic surrogate models, achieving significant gains in QD solution quality under costly and stochastic evaluations.
- Formalized evaluation metrics for UQ in quality-diversity algorithms, quantifying the coverage-uncertainty tradeoff in MAP-Elites archives and enabling principled comparison across surrogate-based QD methods.

Centre National de la Recherche Scientifique (CNRS) | ML Research Intern May 2022 - Jul 2022

Toulouse, France

Detection of pathological oscillations in epilepsy using signal processing and convolutional neural networks.

Supervised by [Dr. Ludovic Gardy](#), [Pr. Emmanuel Barbeau](#), & [Pr. Christophe Hurter](#).

- Accelerated EEG signal and image processing pipelines by 15% via Cython optimization, enabling faster throughput for downstream pathological oscillation analysis.
- Built 7 NoSQL databases for multi-channel EEG storage across macro and micro electrode scales, enabling efficient batch retrieval and parallel processing continuous recordings.
- Trained CNN-based fast-ripple detectors on CWT time-frequency images from macro/micro-electrode EEG, improving classification accuracy by 5% through systematic hyperparameter tuning.
- Applied Grad-CAM to localize high-frequency burst activations within EEG time-frequency maps, providing clinicians with visual explanations anchored to known neurophysiological fast-ripple signatures.

SKILLS

Programming & Scripting: Python, Cython, SQL, NoSQL, LaTeX, Markdown, Bash, Shell

ML/DL Frameworks: PyTorch, JAX, TensorFlow, Scikit-learn, OpenCV, PIL

Scientific Computing & Data: NumPy, Pandas, Polars, SciPy, Matplotlib, Seaborn, Streamlit

MLOps & DevOps: Git, GitHub/GitLab CI/CD, Docker, Singularity, Kubernetes, Azure ML, SkyPilot

Monitoring & Development Tools: Weights & Biases, Pytest, Unittest, Grafana

High-Performance Computing: Triton, SLURM, Faiss

Languages: French (Native), English (Fluent), Spanish (Professional)

PUBLICATIONS

Accepted & Published

- [1] Louis Berthier, Ahmed Shokry, Maxime Moreaud, Guillaume Ramelet, and Eric Moulines. “Adaptive Soft Sensing: A Just-in-Time Approach with Self-Organizing Maps and Online Feature Selection”. In: *2026 IEEE 31st International Conference on Emerging Technologies and Factory Automation (ETFA)*. Accepted, to appear. 2026.
- [2] Louis Berthier, Ahmed Shokry, Eric Moulines, Guillaume Ramelet, and Sylvain Desroziers. “A Framework for Knowledge Discovery in Rubber Mixing Processes through Explainable Boosted Models and Uncertainty Quantification”. In: *35th European Symposium on Computer Aided Process Engineering (ESCAPE 35)*. Vol. 4. Systems and Control Transactions. 2025, pp. 1518–1523. DOI: [10.69997/sct.183525](https://doi.org/10.69997/sct.183525). URL: <https://psecommunity.org/LAPSE:2025.0396>.
- [3] Ludovic Gardy, Jonathan Curot, Luc Valton, Louis Berthier, Emmanuel J. Barbeau, and Christophe Hurter. “Detecting Fast-Ripples on Both Micro- and Macro-Electrodes in Epilepsy: A Wavelet-Based CNN Detector”. In: *Journal of Neuroscience Methods* 415 (2025), p. 110350. ISSN: 0165-0270. DOI: [10.1016/j.jneumeth.2024.110350](https://doi.org/10.1016/j.jneumeth.2024.110350). URL: <https://www.sciencedirect.com/science/article/pii/S0165027024002954>.
- [4] Baptiste Magnier, Ghulam Sakhi Shokouh, Louis Berthier, Marcel Pie, and Adrien Ruggiero. “2DSBG: A 2D Semi Bi-Gaussian Filter Adapted for Adjacent and Multi-Scale Line Feature Detection”. In: *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. 2023, pp. 1–5. DOI: [10.1109/ICASSP49357.2023.10095570](https://doi.org/10.1109/ICASSP49357.2023.10095570).

Preprints & Software

- [5] Louis Berthier, Ahmed Shokry, Maxime Moreaud, Guillaume Ramelet, and Eric Moulines. *torchsom: The Reference PyTorch Library for Self-Organizing Maps*. Preprint, Under Review. 2025. arXiv: [2510.11147](https://arxiv.org/abs/2510.11147) [stat.ML]. URL: <https://arxiv.org/abs/2510.11147>.
- [6] Louis Berthier. *torchsom: The Reference PyTorch Library for Self-Organizing Maps*. Version 1.1.1. Apache 2.0; documentation at <https://opensource.michelin.io/TorchSOM/>. 2025. URL: <https://github.com/michelin/TorchSOM>.

Under Review

- [7] Louis Berthier, Ahmed Shokry, Maxime Moreaud, Guillaume Ramelet, Sylvain Desroziers, and Eric Moulines. “Explainable Offline Process Analytics for Within-Grade Quality Variability in Industrial Rubber Mixing”. Under Review. 2026.
- [8] Louis Berthier, Ahmed Shokry, Maxime Moreaud, Guillaume Ramelet, and Aymeric Dieuleveut. “Self-Organized Conformal Prediction: Reducing Regional Coverage Gaps with Unsupervised Group Discovery”. Under Review. 2026.

In Preparation

- [9] Louis Berthier, Ahmed Shokry, Maxime Moreaud, Guillaume Ramelet, and Eric Moulines. “A Unified Online Framework for Adaptive Soft Sensing in High-Dimensional Batch Processes”. To be Submitted. 2026.
- [10] Mahmoud Hegazy, Louis Berthier, Aymeric Dieuleveut, and Michael I. Jordan. “Beyond Anytime: Mask-Valid Conformal Prediction”. To be Submitted. 2026.

PRESENTATIONS & OUTREACH

A Paper A Week (APAW) | Research Seminar Founder & Organizer | Michelin *Apr 2024 - Present*
Clermont-Ferrand, France

- Founded and lead a bi-monthly research seminar for 20+ researchers and data scientists.
- Facilitated paper discussions bridging ML theory and industrial applications across ML, CV, NLP, and UQ.
- Presented deep dives on: *Attention Is All You Need*, *Quality-Diversity Optimization*, *HyenaDNA*, *Conformal Prediction*, and *TorchSOM*.

Invited Talks & Conference Presentations

Worldwide

- **Michelin DoctoBib’Day** (Clermont-Ferrand, 2026): *A Unified Online Framework for Adaptive Soft Sensing in High-Dimensional Batch Processes*.
- **Mathematical PhD Seminar, École Polytechnique** (Palaiseau, 2025): *TorchSOM: Applications to Online Sensing and Uncertainty Quantification*.
- **ESCAPE 35** (Ghent, Belgium, 2025): *Knowledge Discovery in Large-Scale Batch Processes through Explainable Boosted Models and Uncertainty Quantification: Application to Rubber Mixing*.
- **Data Scientists Network, Michelin** (Clermont-Ferrand, 2025): *Online Sensing for Quality Monitoring*.
- **MT180 National Finalist & Public Award Winner** (Paris, 2025): *Robust real-time quality monitoring and reliable adaptation of manufacturing processes in the tire industry*.

Poster Sessions

Worldwide

- **Local Dynamic Calibration via Self-Organized CP**: SIAM UQ26 (Minneapolis, 2026) and Michelin DoctoBib’Day (Clermont-Ferrand, 2026).
- **Knowledge Discovery in Large-Scale Batch Processes**: Welcome Day IP Paris (2024). Best Poster, Mathematics category.
- **A Framework for Knowledge Discovery in Rubber Mixing Processes**: Michelin Doctoral Day (Clermont-Ferrand, 2024).
- **2DSBG: A 2D Semi Bi-Gaussian Filter for Adjacent and Multi-Scale Line Feature Detection**: ICASSP 2023 (Rhodes, Greece). With A. Ruggiero, M. Pie, G. S. Shokouh, B. Magnier.

TEACHING

Teacher Assistant | ENSTA Paris

Sep 2024 - Oct 2025

Paris, France

- Co-taught lab sessions (20h/year, two consecutive years) for first-year engineers (B3/MSc1) on data preprocessing, model training, and evaluation under [Pr. Philippe Xu](#).

HONOURS & AWARDS

Honours

- **Master Thesis with Distinction (1st)** | Imperial College London, 2023
Proposal for a PhD in the AIRL laboratory under the supervision of [Pr. Antoine Cully](#).
- **Master of Science with Merit (2:1)** | Imperial College London, 2023
- **Diplôme d'ingénieur with First Class Honours (1st)** | IMT Mines Alès, 2023
Jury's congratulations, ranked 2nd in the CS & AI department, and in the top 5% of all students.

Grants & Academic Awards

- **MT180 National Finalist & Public Award (€100)** | Institut Polytechnique de Paris, 2025
- **Best PhD Poster Award (€500)** | Institut Polytechnique de Paris, 2024
- **International Mobility Scholarship (€4000)** | Région Occitanie, 2022
- **Alumni Scholarship (€1300)** | Mines Alès Alumni, 2022
- **Excellence Scholarship (€4000)** | IMT Mines Alès, 2022

Competitions & Hackathons

- **2nd Place (of 25 teams), Hack the World(s)** | Paris, 2026
Built a hierarchical JEPA world model on the complete fly-brain connectome with team *Piaget*; reposted and congratulated by [Yann LeCun](#). [\[Project\]](#)
- **Selected Participant, Mistral Worldwide Hackathon** | Paris, 2026
Selected from 1,700+ applicants; built *Cerno*, an AI recruitment platform with adaptive voice-AI interviews, in 36 hours. [\[Video\]](#)
- **Paris Fintech Hackathon (Solve with AI)** | HEC Paris, 2026
Built *WEBI*, an AI copilot for structured bond products (Gemini 2.0 Flash + FAISS). [\[Devpost\]](#)
- **Winner, Manufacturing IT Hackathon AI'NNOV (€500)** | Michelin, 2024
- **Creativity Laureate, Communication & Media (€80)** | Cora Hypermarket, 2021

VOLUNTEERING

EmaVisual (Photography & Video Association) | Vice President

Apr 2021 - Apr 2022

Alès, France

- Led a team of 10 photographers to cover 25 university events.
- Edited and produced multiple videos and over 10,000 photos.

Bureau des Sports (Sports Association) | Vice President

Feb 2021 - Apr 2022

Alès, France

- Led a 25-person team to manage 5 tournaments and oversee 20 sports.
- Secured partnerships with major brands, including Armor Lux and Tony Parker.
- Managed the association's website and IT infrastructure.

Tsiky Zanaka (Humanitarian Association) | Tsikyst Volunteer

Oct 2020 - Oct 2021

Alès, France

- Raised over €3500 to fund and support solidarity projects in Madagascar.
- Organized and participated in fundraising events, including cooking for 300 students.

University | Handball Player

Oct 2020 - Nov 2023

FR & UK

- Achieved 3rd place in the highest national university league (BUCS, UK).
- Achieved 4th place in the highest national university league (FFSU, France).
- Winner of the 49th Challenge Centrale Lyon
- Winner of the 19th TRAMS